

Benjamin Yu Sheng Chang

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EDUCATION

University of Toronto

Bachelor's of Science, Computer Science and Mathematics

Toronto, On

Sep 2022 — Present

- Cumulative GPA: 3.75/4.0 | Dean's List $\times$ 3
- Teaching Assistant: Linear Algebra I (Winter 2026) Held weekly office hours and graded assignments for a 500+ class
- Relevant Coursework: Machine Learning, Spectral Graph Theory, Data Structure, Algorithm Design, Software Design

WORK EXPERIENCE

Machine Learning Research Assistant | *Acceleration Consortium*

Jan 2026 — Apr 2026

- Architected an end-to-end ML workflow using **MLflow** and **AWS ECS**, automating continuous model deployment
- Developed **Pytorch** surrogate models for Bayesian optimization to accelerate material science discovery for self-driving lab
- Stabilized model training by leveraging more informative structure of data representation in limited-data condition
- Rewrote Python notebooks into a maintainable research codebase, enabling integration with industry applications
- Led architectural decisions that connected disparate research and engineering modules into a maintainable system.

Full Stack Developer Intern | *Data On The Spot*

Jan 2025 — Aug 2025

- Built a backend architecture for real-time collaborative text editors with a dynamic rule system in TypeScript and NoSQL, scalable to **1000+** concurrent users, which enabled the company to secure its largest potential client
- Deployed a internal Rust app in Tauri with a COM library to connect physical voting hardware, supporting **2000+** keypads simultaneously for hybrid voting sessions
- Implemented an AI-powered email automation workflow in Make.com using external APIs to optimize customer retention and streamline follow-up processes
- Designed and implemented a full registration system with a dynamic form builder and integrated logic engine in **TypeScript**, **Drizzle**, and **PostgreSQL**, enabling customizable workflows and adaptive user experiences

RESEARCH EXPERIENCE

Physical Informed Neural Network Under Limited Data

Glasgow, UK

Supervisors: Professor Eduardo Patelli

May 2026 — Aug 2026(Expected)

- Research focused on improving state-of-the-art PINN methods under limited-data conditions

PROJECTS

Predictive Modeling for Crypto Horse Racing | *Python, Pytorch, imblearn*

- Built **regression** and **NN** models with feature engineering and data balancing to predict race winners ($R^2 = 0.67$)
- Applied **SMOTE** to balance winner-loser classes and trained a neural network model leveraging horse history, traits, and in-game metrics to predict race outcomes

Fullhouse Hackathon | *Python, Pytorch, Numpy* - <https://github.com/Benjamin-Yu-Sheng-Chang/fullhouse-hackathon>

- Placed in top 100 qualifier in first quantitative poker Hackathon in UK
- Built a heuristic learning loop using the **DeepSeek R1** model to iteratively improve poker bot strategies, with rigorous benchmarking of bot performance and comprehensive correctness validation.

QNN and QAOA Algorithm | *Qiskit, Python, Pytorch, sQUlearn*

- Implemented **QAOA** with Qiskit to solve a classical combinatorial optimization problem
- Built **QNN regression** with multiclass quantum kernel methods utilizing sQUlearn
- Built and trained a variational quantum classifier from scratch with Qiskit Machine Learning

SKILLS

- **Programming Languages:** Python, C/C++, JS/TS, HTML/CSS, SQL, Rust, Latex/Typst
- **Frameworks:** Next/React, Svelte, Node.js, MLflow
- **Technologies:** Docker, Github actions, AWS
- **AI agents:** Claude Code, Codex